FOCUS 3 • PHYSICAL CHEMISTRY • 物理化学

# The Second and Third Laws (continued)

## 热力学第二、第三定律（续）

Topics 3B · 3C

Lesson 2

WHERE WE ARE

# Recap of Lesson 1

- Entropy S is a state function: $dS = dq_{rev}/T$.
  熵是状态函数
- ΔS_total ≥ 0 for spontaneous change (Second Law).
  第二定律

· The Second Law decides direction; now we make it practical.

# What we will cover

- 3B Entropy Changes Accompanying Specific Processes.
  特定过程的熵变
- 3C The Third Law and absolute entropies.
  第三定律

TOPIC 3B

# Entropy Changes Accompanying Specific Processes

特定过程的熵变

1. Expansion · 膨胀
2. Phase transitions · 相变
3. Heating · 加热
4. Composite processes · 复合过程

➤ Why do you need to know this material?

The changes in entropy accompanying a variety of basic physical processes occur throughout the application of the Second Law to chemistry.

➤ What is the key idea?

The change in entropy accompanying a process is calculated by identifying a reversible path between the initial and final states.

# Entropy change

$$dS = \frac{dq_{rev}}{T} \qquad \Delta S = \int_i^f \frac{dq_{rev}}{T}$$ Entropy change [definition] (3B.1a)

$$\Delta S_{sur} = \frac{q_{sur}}{T_{sur}}$$ Entropy change of surroundings (3B.1b)

$$\Delta S_{tot} = \Delta S + \Delta S_{sur}$$ Total entropy change (3B.1c)

- A spontaneous process is also irreversible (in the thermodynamic sense) and a process for which $\Delta S_{tot} = 0$ is at equilibrium.

# Expansion

$$\Delta S = nR\ln\frac{V_f}{V_i} \qquad \text{(3B.2)}$$

Entropy change for the isothermal expansion of a perfect gas

- The total change in entropy, however, does depend on how the expansion takes place.

$dq_{sur} = -dq$. For the reversible isothermal expansion of a perfect gas $q_{rev} = nRT\ln(V_f/V_i)$, so $q_{sur} = -nRT\ln(V_f/V_i)$, and consequently

$$\Delta S_{sur} = -\frac{q_{rev}}{T} = -nR\ln\frac{V_f}{V_i} \qquad \text{(3B.3a)}$$

$\Delta S_{tot} = 0$

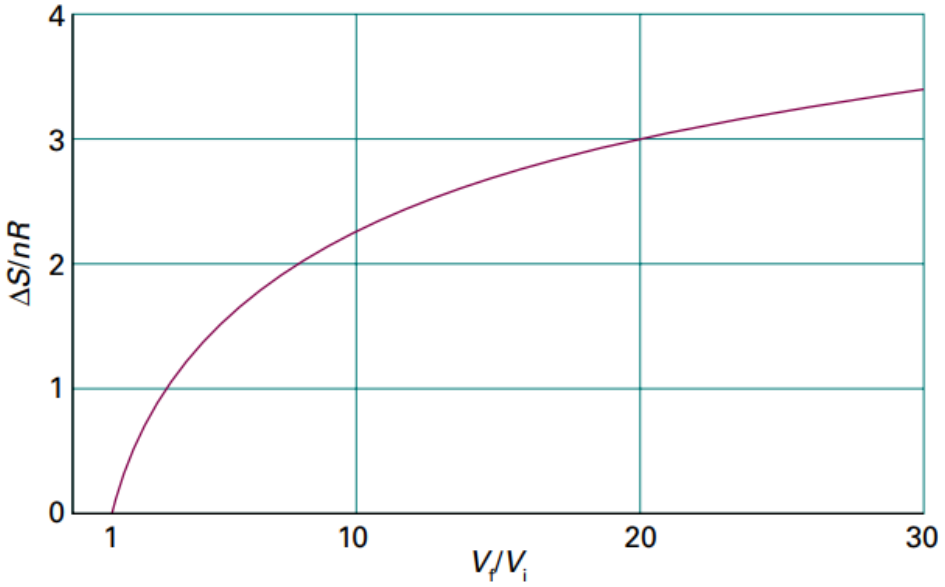



**Figure 3B.1** The logarithmic increase in entropy of a perfect gas as it expands isothermally.

# Expansion

| 过程 | 系统熵变 ΔSsys | 总熵变（系统 + 环境）$\Delta S_{tot}$ |
|---|---|---|
| 可逆 | 由初末态决定，$\Delta S=\int dq_{rev}/T$ | $\Delta S_{tot}=0$ |
| 不可逆 | 和同初末态可逆路径的熵变相等 | $\Delta S_{tot}>0$ |

$$\Delta S = nR\ln\frac{V_f}{V_i}$$

Entropy change for the isothermal expansion of a perfect gas (3B.2)

- The isothermal expansion occurs freely (if the expansion is into a vacuum) no work is done (w = 0).

$\Delta U = 0$, $\Delta U = q + w$, $q = 0$, $q_{sur} = 0$ and hence $\Delta S_{sur} = 0$

$$\Delta S_{tot} = nR\ln\frac{V_f}{V_i}$$

$\Delta S_{tot} > 0$, irreversible process

$S_{tot}$ is not a state function

**Brief illustration 3B.1**

When the volume of any perfect gas is doubled at constant temperature, $V_f/V_i = 2$, and hence the change in molar entropy of the system is

$$\Delta S_m = (8.3145\,\mathrm{J\,K^{-1}\,mol^{-1}}) \times \ln 2 = +5.76\,\mathrm{J\,K^{-1}\,mol^{-1}}$$

If the change is carried out reversibly, the change in entropy of the surroundings is $-5.76\,\mathrm{J\,K^{-1}\,mol^{-1}}$ (the 'per mole' meaning per mole of gas molecules in the sample). The total change in entropy is 0. If the expansion is free, the change in molar entropy of the gas is still $+5.76\,\mathrm{J\,K^{-1}\,mol^{-1}}$, but that of the surroundings is 0, and the total change is $+5.76\,\mathrm{J\,K^{-1}\,mol^{-1}}$.

# Reversible process and Irreversible process

- 可逆过程：过程发生后，可以让系统和外界，全部同时回到原来初始状态，不留任何痕迹
- 不可逆过程：一旦发生，就算系统复原了，外界一定留下无法消除的变化（痕迹），不能全部复原。
- 有温差传热、有摩擦、突然变化、扩散混合 → 不可逆
- 无限慢、无耗散、无限小压差 / 温差 → 可逆（理想）

- 不可逆过程：

1. 存在有限大小温差的传热（高温物体直接向低温物体传热）
2. 有摩擦、粘滞阻力、湍流（机械能耗散变热）
3. 自由膨胀（气体突然冲开阀门向真空膨胀）
4. 气体快速膨胀 / 压缩（系统压强和外压相差很大，不是无限缓慢）
5. 扩散、混合、化学反应自发进行

# Phase transition相变

- Normal transition temperature正常相变温度, $T_{trs}$, the temperature at which two phases are in equilibrium at 1 atm
- At $T_{trs}$, any transfer of energy as heat between the system and its surroundings is reversible because the two phases in the system are in equilibrium.
- At a phase transition, T is constant and the heat is the transition enthalpy. $q = \Delta H_{trs}$

$$\Delta_{trs}S = \frac{\Delta_{trs}H}{T_{trs}}$$

Entropy of phase transition [at $T_{trs}$] (3B.4)

- Exothermic, $\Delta_{trs}H < 0$, freezing or condensing, $\Delta_{trs}S < 0$
- Endothermic, $\Delta_{trs}H > 0$, Melting and vaporization, $\Delta_{trs}S > 0$

$\Delta S_{tot} = 0$

· At the transition temperature, heating is a reversible process.

# Phase transition

- A wide range of liquids give approximately the same standard entropy of vaporization (about 85 J $K^{-1}$ $mol^{-1}$): this empirical observation is called Trouton's rule特鲁顿规则.
- A similar change in volume occurs when any liquid evaporates and becomes a gas. Hence, all liquids can be expected to have similar standard entropies of vaporization
- Liquids that show significant deviations from Trouton's rule: strong molecular interactions ($H_2O$), fewer translational and rotational states are accessible for molecules with low mass and moments of inertia惯性 ($CH_4$)

**Table 3B.2** The standard enthalpies and entropies of vaporization of liquids at their boiling temperatures*

| | $\Delta_{vap}H^{\ominus}$/(kJ $mol^{-1}$) | $\theta_b$/°C | $\Delta_{vap}S^{\ominus}$/(J $K^{-1}$ $mol^{-1}$) |
|---|---|---|---|
| Benzene | 30.8 | 80.1 | 87.2 |
| Carbon tetrachloride | 30 | 76.7 | 85.8 |
| Cyclohexane | 30.1 | 80.7 | 85.1 |
| Hydrogen sulfide | 18.7 | –60.4 | 87.9 |
| Methane | 8.18 | –161.5 | 73.2 |
| Water | 40.7 | 100.0 | 109.1 |

# Phase transition

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**Brief illustration 3B.2**

There is no hydrogen bonding in liquid bromine and $Br_2$ is a heavy molecule which is unlikely to display unusual behaviour in the gas phase, so it is safe to use Trouton's rule. To predict the standard molar enthalpy of vaporization of bromine given that it boils at 59.2 °C, use Trouton's rule in the form

$$\Delta_{vap}H^{\ominus} = T_b \times (85\,J\,K^{-1}\,mol^{-1})$$

Substitution of the data then gives

$$\Delta_{vap}H^{\ominus} = (332.4\,K) \times (85\,J\,K^{-1}\,mol^{-1})$$
$$= +2.8 \times 10^4\,J\,mol^{-1} = +28\,kJ\,mol^{-1}$$

The experimental value is +29.45 kJ $mol^{-1}$.

# Heating

$$dS = \frac{dq_{rev}}{T} \qquad \Delta S = \int_i^f \frac{dq_{rev}}{T}$$

Entropy change [definition] (3B.1a)

$$S(T_f) = S(T_i) + \int_{T_i}^{T_f} \frac{dq_{rev}}{T}$$

- At constant pressure, $dq_{rev} = dH = C_p dT$.

$$S(T_f) = S(T_i) + \int_{T_i}^{T_f} \frac{C_p dT}{T}$$

Entropy variation with temperature [constant $p$] (3B.6)

- If Cp is constant, $S(T_f) = S(T_i) + C_p \int_{T_i}^{T_f} \frac{dT}{T} = S(T_i) + C_p \ln \frac{T_f}{T_i}$
- At constant volumn, use $C_v$

**Brief illustration 3B.3**

The molar constant-volume heat capacity of water at 298 K is $75.3\,J\,K^{-1}\,mol^{-1}$. The change in molar entropy when it is heated from 20 °C (293 K) to 50 °C (323 K), supposing the heat capacity to be constant in that range, is therefore

$$\Delta S_m = S_m(323K) - S_m(293K) = (75.3\,JK^{-1}\,mol^{-1}) \times \ln \frac{323K}{293K}$$

$$= +7.34\,JK^{-1}\,mol^{-1}$$

· Heating a substance always increases its entropy.

$$\Delta S = nR\ln\frac{V_f}{V_i}$$

Entropy change for the isothermal expansion of a perfect gas (3B.2)

# Composite processes复杂过程

- It might be convenient to split the path into two steps: an isothermal expansion to the final volume, followed by heating at constant volume to the final temperature.

**Example 3B.1** Calculating the entropy change for a composite process

Calculate the entropy change when argon at 25 °C and 1.00 bar in a container of volume 0.500 $dm^3$ is allowed to expand to 1.000 $dm^3$ and is simultaneously heated to 100 °C. (Take the molar heat capacity at constant volume to be $\frac{3}{2}R$.)

***Collect your thoughts*** As remarked in the text, you can break the overall process down into two steps: isothermal expansion to the final volume, followed by heating at constant volume to the final temperature. The entropy change in the first step is given by eqn 3B.2 and that of the second step, provided $C_V$ is independent of temperature, by eqn 3B.7 (with $C_V$ in place of $C_p$). In each case you need to know $n$, the amount of gas molecules, which can be calculated from the perfect gas equation and the data for the initial state by using $n = p_i V_i / RT_i$.

# Composite processes复杂过程

$$\Delta S = nR\ln\frac{V_f}{V_i} \quad \text{(3B.2)}$$

Entropy change for the isothermal expansion of a perfect gas

$$S(T_f) = S(T_i) + \int_{T_i}^{T_f}\frac{C_p\,dT}{T} \quad \text{(3B.6)}$$

Entropy variation with temperature [constant $p$]

- It might be convenient to split the path into two steps: an isothermal expansion to the final volume, followed by heating at constant volume to the final temperature.

***The solution*** The amount of gas molecules is

$$n = \frac{(1.00\times10^5\,\text{Pa})\times(0.500\times10^{-3}\,\text{m}^3)}{(8.3145\,\text{JK}^{-1}\,\text{mol}^{-1})\times298\,\text{K}} = 0.0201\ldots\text{mol}$$

From eqn 3B.2 the entropy change in the isothermal expansion from $V_i$ to $V_f$ is

$$\begin{aligned}\Delta S(\text{Step 1}) &= nR\ln\frac{V_f}{V_i}\\ &= 0.0201\ldots\text{mol}\times(8.3145\,\text{JK}^{-1}\,\text{mol}^{-1})\ln\frac{1.000\,\text{dm}^3}{0.500\,\text{dm}^3}\\ &= +0.116\ldots\text{JK}^{-1}\end{aligned}$$

From eqn 3B.6, the entropy change in the second step, heating from $T_i$ to $T_f$ at constant volume, is

$$\begin{aligned}\Delta S(\text{Step 2}) &= nC_{V,m}\ln\frac{T_f}{T_i} = \tfrac{3}{2}nR\ln\frac{T_f}{T_i}\\ &= \tfrac{3}{2}\times(0.0201\ldots\text{mol})\times(8.3145\,\text{JK}^{-1}\,\text{mol}^{-1})\ln\frac{373\,\text{K}}{298\,\text{K}}\\ &= +0.0564\ldots\text{JK}^{-1}\end{aligned}$$

The overall entropy change of the system, the sum of these two changes, is

$$\Delta S = 0.116\ldots\text{JK}^{-1} + 0.0564\ldots\text{JK}^{-1} = +0.173\,\text{JK}^{-1}$$

# Self-test

$$\Delta S = nR\ln\frac{V_f}{V_i}$$ Entropy change for the isothermal expansion of a perfect gas (3B.2)

$$S(T_f) = S(T_i) + \int_{T_i}^{T_f}\frac{C_p\,dT}{T}$$ Entropy variation with temperature [constant $p$] (3B.6)

- Calculate the entropy change when the same initial sample is compressed to 0.0500 $dm^3$ and cooled to −25 °C.

Entropy change for ideal gas with change in $T$ and $V$:

$$\Delta S = nC_{V,m}\ln\frac{T_2}{T_1} + nR\ln\frac{V_2}{V_1}, \quad C_{V,m} = \tfrac{3}{2}R$$

$$T_1 = 298.15\ \mathrm{K},\ T_2 = 248.15\ \mathrm{K},\ V_1 = 0.500\ \mathrm{dm}^3,\ V_2 = 0.0500\ \mathrm{dm}^3$$

$$\boldsymbol{\Delta S = -0.432\ \mathrm{J\ K^{-1}}}$$

- 被压缩（体积缩小，有序度上升，熵减）+降温（分子热运动减弱，熵减），系统总熵减少

TOPIC 3B · SELF-TEST

# Check your understanding

**For an isothermal expansion of a perfect gas, ΔS is:**

- A. negative
- B. zero
- C. positive
- D. undefined

**Answer · 答案: C — expansion into a larger volume increases entropy ($\Delta S = nR \ln(V_f/V_i) > 0$).**

*The gas disperses into a larger volume.*

## Checklist of concepts

- ☐ 1. The entropy of a perfect gas increases when it expands isothermally.
- ☐ 2. The change in entropy of a substance accompanying a change of state at its transition temperature is calculated from its enthalpy of transition.
- ☐ 3. The increase in entropy when a substance is heated is calculated from its heat capacity.

## Checklist of equations

| Property | Equation | Comment | Equation number |
|---|---|---|---|
| Entropy of isothermal expansion | $\Delta S = nR \ln(V_f/V_i)$ | Perfect gas | 3B.2 |
| Entropy of transition | $\Delta_{trs}S = \Delta_{trs}H/T_{trs}$ | At the transition temperature | 3B.4 |
| Variation of entropy with temperature | $S(T_f) = S(T_i) + C \ln(T_f/T_i)$ | The heat capacity, $C$, is independent of temperature and no phase transitions occur; $C = C_p$ for constant pressure and $C_V$ for constant volume. | 3B.7 |

*These three cover most entropy calculations in chemistry.*

FOCUS 3 · QUESTION

# How do we make spontaneity practical?

**Think about it · 想一想**

**Computing $\Delta S_{total}$ needs both the system and the surroundings. How could we decide spontaneity using only the system's own properties?**

*Define a function that combines enthalpy and entropy — that is the Gibbs energy.*

TOPIC 3C

# The Measurement of Entropy

熵的测量

1. Calorimetric measurement · 量热测量
2. The Third Law · 第三定律
3. Absolute entropies · 绝对熵

➤ Why do you need to know this material?

For entropy to be a quantitatively useful concept it is important to be able to measure it: the calorimetric procedure is described here. The Third Law of thermodynamics is used to report the measured values.

➤ What is the key idea?

The entropy of a perfectly crystalline solid is zero at $T = 0$.

# Measuring entropy by calorimetry

$$S(T_2) = S(T_1) + \int_{T_1}^{T_2} \frac{C_p(T)}{T} \mathrm{d}T$$ Entropy and temperature (3C.1a)

$$\Delta S(T_{\mathrm{trs}}) = \frac{\Delta_{\mathrm{trs}} H(T_{\mathrm{trs}})}{T_{\mathrm{trs}}}$$ Entropy of phase transition (3C.1b)

$$S_{\mathrm{m}}(T) = S_{\mathrm{m}}(0) + \overbrace{\int_0^{T_{\mathrm{f}}} \frac{C_{p,\mathrm{m}}(\mathrm{s},T')}{T'} \mathrm{d}T'}^{\text{Heat solid to its melting point}} + \overbrace{\frac{\Delta_{\mathrm{fus}} H}{T_{\mathrm{f}}}}^{\text{Entropy of fusion}} + \overbrace{\int_{T_{\mathrm{f}}}^{T_{\mathrm{b}}} \frac{C_{p,\mathrm{m}}(\mathrm{l},T')}{T'} \mathrm{d}T'}^{\text{Heat liquid to its boiling point}} + \overbrace{\frac{\Delta_{\mathrm{vap}} H}{T_{\mathrm{b}}}}^{\text{Entropy of vaporization}} + \overbrace{\int_{T_{\mathrm{b}}}^{T} \frac{C_{p,\mathrm{m}}(\mathrm{g},T')}{T'} \mathrm{d}T'}^{\text{Heat vapour to the final temperature}}$$

# Measuring entropy by calorimetry

- The area under C/T versus T is the entropy increment.
  C/T–T 面积
- Add phase-transition entropy at each transition.
- The difficulty of measuring heat capacities near T = 0.

The heat capacity of a non-metallic solid is proportional to $T^3$ when T is low, Debye extrapolation德拜外推 (or the Debye $T^3$ law)

- Cp is measured down to as low a temperature as possible and a curve of the form $aT^3$ is fitted to the data.

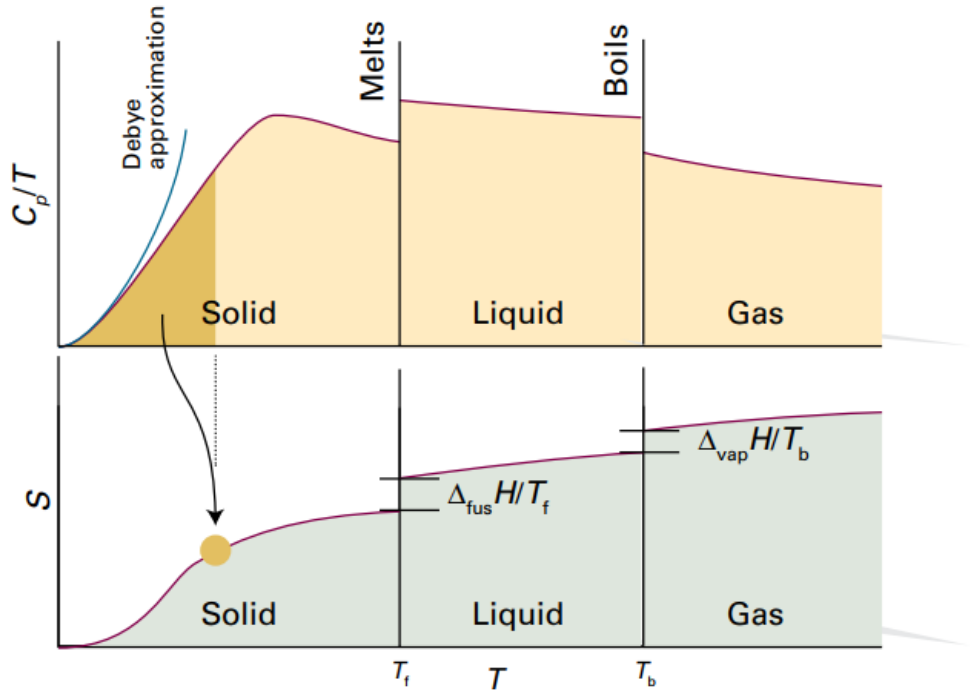



**Figure 3C.1** The variation of $C_p/T$ with the temperature for a sample is used to evaluate the entropy, which is equal to the area beneath the upper curve up to the corresponding temperature, plus the entropy of each phase transition encountered between $T = 0$ and the temperature of interest. For instance, the entropy denoted by the yellow dot on the lower curve is given by the dark shaded area in the upper graph.

· Low-temperature heat-capacity data give absolute entropies.

# Measuring entropy by calorimetry

**Brief illustration 3C.1**

The standard molar entropy of nitrogen gas at 25 °C has been calculated from the following data:

| | Contribution to $S_m^{\ominus}/(J\,K^{-1}\,mol^{-1})$ |
|---|---|
| Debye extrapolation | 1.92 |
| Integration, from 10 K to 35.61 K | 25.25 |
| Phase transition at 35.61 K | 6.43 |
| Integration, from 35.61 K to 63.14 K | 23.38 |
| Fusion at 63.14 K | 11.42 |
| Integration, from 63.14 K to 77.32 K | 11.41 |
| Vaporization at 77.32 K | 72.13 |
| Integration, from 77.32 K to 298.15 K | 39.20 |
| Correction for gas imperfection | 0.92 |
| Total | 192.06 |

Therefore, $S_m^{\ominus}(298.15\,K) = S_m(0) + 192.1\,J\,K^{-1}\,mol^{-1}$. The Debye extrapolation is explained in the next paragraph.

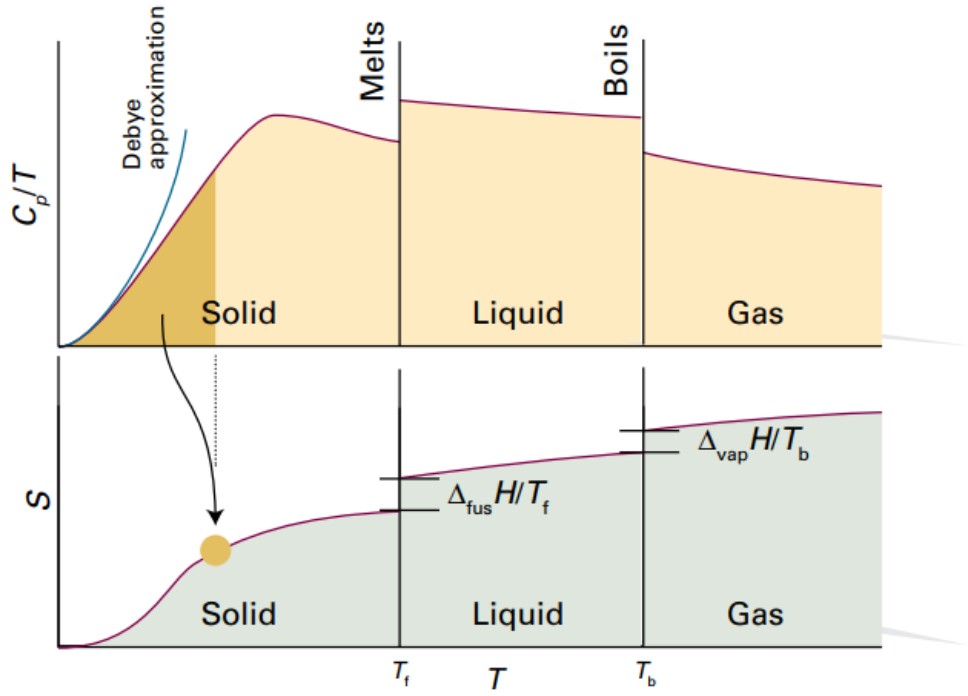



**Figure 3C.1** The variation of $C_p/T$ with the temperature for a sample is used to evaluate the entropy, which is equal to the area beneath the upper curve up to the corresponding temperature, plus the entropy of each phase transition encountered between $T = 0$ and the temperature of interest. For instance, the entropy denoted by the yellow dot on the lower curve is given by the dark shaded area in the upper graph.

# Calculating the entropy at low temperatures

The molar constant-pressure heat capacity of a certain non-metallic solid at 4.2 K is $0.43\,\mathrm{J\,K^{-1}\,mol^{-1}}$. What is its molar entropy at that temperature?

***Collect your thoughts*** Because the temperature is so low, you can assume that the heat capacity varies with temperature according to $C_{p,\mathrm{m}}(T) = aT^3$, in which case you can use eqn 3C.1a to calculate the entropy at a temperature $T$ in terms of the entropy at $T = 0$ and the constant $a$. When the integration is carried out, it turns out that the result can be expressed in terms of the heat capacity at the temperature $T$, so the data can be used directly to calculate the entropy.

***The solution*** The integration required is

$$S_{\mathrm{m}}(T) = S_{\mathrm{m}}(0) + \int_0^T \frac{aT'^{\,3}}{T'}\,\mathrm{d}T' = S_{\mathrm{m}}(0) + a\overbrace{\int_0^T T'^{\,2}\,\mathrm{d}T'}^{\text{Integral A.1}}$$
$$= S_{\mathrm{m}}(0) + \tfrac{1}{3}aT^3 = S_{\mathrm{m}}(0) + \tfrac{1}{3}C_{p,\mathrm{m}}(T)$$

from which it follows that

$$S_{\mathrm{m}}(4.2\,\mathrm{K}) = S_{\mathrm{m}}(0) + 0.14\,\mathrm{J\,K^{-1}\,mol^{-1}}$$

# Calculating the entropy at low temperatures

***Self-test 3C.1*** For metals, there is also a contribution to the heat capacity from the electrons which is linearly proportional to $T$ when the temperature is low; that is, $C_{p,\mathrm{m}}(T) = bT$. Evaluate its contribution to the entropy at low temperatures.

$$\begin{aligned} S_{\mathrm{m}}(T) &= S_{\mathrm{m}}(0) + \int_0^T \frac{bT'}{T'}\,\mathrm{d}T' \\ &= S_{\mathrm{m}}(0) + \int_0^T b\,\mathrm{d}T' \\ &= \boldsymbol{S_{\mathrm{m}}(0) + bT} \end{aligned}$$

# The Third Law

- At T = 0, all energy of thermal motion has been quenched, and in a perfect crystal all the atoms or ions are in a regular, uniform array.
- There is only one way of arranging the molecules when they are all in the ground state, which is the case at T = 0. W = 1, S = k lnW = 0.
- Nernst heat theorem能斯特热定理 The entropy change accompanying any physical or chemical transformation approaches zero as the temperature approaches zero: ΔS → 0 as T → 0 provided all the substances involved are perfectly ordered.

**Brief illustration 3C.2**

The entropy of the transition between orthorhombic sulfur, α, and monoclinic sulfur, β, can be calculated from the transition enthalpy (402 J mol$^{-1}$) at the transition temperature (369 K):

$$\Delta_{trs}S(369\,K) = S_m(\beta,369\,K) - S_m(\alpha,369\,K)$$
$$= \frac{\Delta_{trs}H}{T_{trs}} = \frac{402\,J\,mol^{-1}}{369\,K} = 1.09\,J\,K^{-1}\,mol^{-1}$$

The entropies of the α and β allotropes can also be determined by measuring their heat capacities from $T = 0$ up to $T = 369$ K. It is found that $S_m(\alpha,369\,K) = S_m(\alpha,0) + 37\,J\,K^{-1}\,mol^{-1}$ and $S_m(\beta,369\,K) = S_m(\beta,0) + 38\,J\,K^{-1}\,mol^{-1}$. These two values imply that at the transition temperature

$$\Delta_{trs}S(369\,K) = \{S_m(\beta,0) + 38\,J\,K^{-1}\,mol^{-1}\} - \{S_m(\alpha,0) + 37\,J\,K^{-1}\,mol^{-1}\}$$
$$= S_m(\beta,0) - S_m(\alpha,0) + 1\,J\,K^{-1}\,mol^{-1}$$

On comparing this value with the one above, it follows that $S_m(\beta,0) - S_m(\alpha,0) \approx 0$, in accord with the theorem.

· The Third Law lets us assign absolute entropies, not just changes.

# The Third Law

- Third Law of thermodynamics热力学第三定律The entropy of all perfect crystalline substances is zero at T = 0
- In certain cases W > 1 at T = 0 and therefore S(0) > 0
- If S(0) > 0 the substance is said to have a residual entropy残余熵.

Ice has a residual entropy of 3.4 J K−1 mol−1. It stems from the arrangement of the hydrogen bonds between neighbouring water molecules: a given O atom has two short O−H bonds and two long O…H bonds to its neighbours, but there is a degree of randomness in which two bonds are short and which two are long

# Third-Law entropies

- Entropies reported on the basis that S(0) = 0 are called Third-Law entropies (and commonly just 'entropies')
- The standard reaction entropy

$$\Delta_r S^{\ominus} = \sum_{\text{Products}} \nu S_m^{\ominus} - \sum_{\text{Reactants}} \nu S_m^{\ominus}$$

Standard reaction entropy [definition] (3C.3a)

$$\Delta_r S^{\ominus} = \sum_J \nu_J S_m^{\ominus}(J)$$

(3C.3b)

where the $\nu_J$ are signed (+ for products, − for reactants) stoichiometric numbers

- Standard reaction entropies are likely to be positive if there is a net formation of gas in a reaction, and are likely to be negative if there is a net consumption of gas.

# Third-Law entropies

**Brief illustration 3C.3**

To calculate the standard reaction entropy of $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ at 298 K, use the data in Table 2C.4 of the *Resource section* to write

$$\begin{aligned}\Delta_r S^\ominus &= S_m^\ominus(H_2O,l) - \{S_m^\ominus(H_2,g) + \tfrac{1}{2}S_m^\ominus(O_2,g)\} \\ &= 69.9\,J\,K^{-1}\,mol^{-1} - \{130.7 + \tfrac{1}{2}(205.1)\}\,J\,K^{-1}\,mol^{-1} \\ &= -163.4\,J\,K^{-1}\,mol^{-1}\end{aligned}$$

The negative value is consistent with the conversion of two gases to a compact liquid.

# Third-Law entropies

- Solutions of cations cannot be prepared in the absence of anions

$$S^{\ominus}(H^+,aq) = 0 \qquad \text{Ions in solution [convention]} \qquad (3C.4)$$

**Brief illustration 3C.4**

The standard molar entropy of $Cl^-(aq)$ is $+57\,J\,K^{-1}\,mol^{-1}$ and that of $Mg^{2+}(aq)$ is $-128\,J\,K^{-1}\,mol^{-1}$. That is, the molar entropy of $Cl^-(aq)$ is $57\,J\,K^{-1}\,mol^{-1}$ higher than that of the proton in water (presumably because it induces less local structure in the surrounding water), whereas that of $Mg^{2+}(aq)$ is $128\,J\,K^{-1}\,mol^{-1}$ lower (presumably because its higher charge induces more local structure in the surrounding water).

# The temperature dependence of reaction entropy

$$S(T_2) = S(T_1) + \int_{T_1}^{T_2} \frac{C_p(T)}{T} \mathrm{d}T$$ Entropy and temperature (3C.1a)

$$S_{\mathrm{m}}(T_2) = S_{\mathrm{m}}(T_1) + \int_{T_1}^{T_2} \frac{C_{p,\mathrm{m}}(T)}{T} \mathrm{d}T$$

$$\Delta_{\mathrm{r}}S^{\ominus}(T_2) = \Delta_{\mathrm{r}}S^{\ominus}(T_1) + \int_{T_1}^{T_2} \frac{\Delta_{\mathrm{r}}C_p^{\ominus}}{T} \mathrm{d}T \qquad (3C.5a)$$

- $\Delta_rC_p^{\ominus}$ is the difference of the molar heat capacities of products and reactants under standard conditions weighted by the stoichiometric numbers

$$\Delta_{\mathrm{r}}C_p^{\ominus} = \sum_{\mathrm{J}} \nu_{\mathrm{J}} C_{p,\mathrm{m}}^{\ominus}(\mathrm{J}) \qquad (3C.5b)$$

- If $\Delta_rC_p^{\ominus}$ is independent of temperature in the range $T_1$ to $T_2$

$$\Delta_{\mathrm{r}}S^{\ominus}(T_2) = \Delta_{\mathrm{r}}S^{\ominus}(T_1) + \Delta_{\mathrm{r}}C_p^{\ominus} \ln \frac{T_2}{T_1} \qquad (3C.5c)$$

# The temperature dependence of reaction entropy

**Brief illustration 3C.5**

The standard reaction entropy for $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$ at 298 K is $-44.42\,J\,K^{-1}\,mol^{-1}$, and the molar heat capacities at constant pressure of the molecules are $H_2O(g)$: $33.58\,J\,K^{-1}\,mol^{-1}$; $H_2(g)$: $28.84\,J\,K^{-1}\,mol^{-1}$; $O_2(g)$: $29.37\,J\,K^{-1}\,mol^{-1}$. It follows that

$$\begin{aligned}\Delta_r C_p^{\ominus} &= C_{p,m}^{\ominus}(H_2O,g) - C_{p,m}^{\ominus}(H_2,g) - \tfrac{1}{2}C_{p,m}^{\ominus}(O_2,g) \\ &= -9.94\,J\,K^{-1}\,mol^{-1}\end{aligned}$$

This value of $\Delta_r C_p^{\ominus}$ is used in eqn 3C.5c to find $\Delta_r S^{\ominus}$ at another temperature, for example at 373 K

$$\begin{aligned}\Delta_r S^{\ominus}(373\,K) &= -44.42\,J\,K^{-1}\,mol^{-1} + (-9.94\,J\,K^{-1}\,mol^{-1}) \times \ln\frac{373\,K}{298\,K} \\ &= -46.65\,J\,K^{-1}\,mol^{-1}\end{aligned}$$

## Checklist of concepts

- ☐ 1. Entropies are determined calorimetrically by measuring the heat capacity of a substance from low temperatures up to the temperature of interest and taking into account any phase transitions in that range.
- ☐ 2. The **Debye extrapolation** (or the *Debye $T^3$-law*) is used to estimate heat capacities of non-metallic solids close to $T = 0$.
- ☐ 3. The **Nernst heat theorem** states that the entropy change accompanying any physical or chemical transformation approaches zero as the temperature approaches zero: $\Delta S \to 0$ as $T \to 0$ provided all the substances involved are perfectly ordered.
- ☐ 4. The **Third Law of thermodynamics** states that the entropy of all perfect crystalline substances is zero at $T = 0$.
- ☐ 5. The **residual entropy** of a solid is the entropy arising from disorder that persists at $T = 0$.
- ☐ 6. **Third-law entropies** are entropies based on $S(0) = 0$.
- ☐ 7. The **standard entropies of ions in solution** are based on setting $S^{\ominus}(H^+,aq) = 0$ at all temperatures.
- ☐ 8. The **standard reaction entropy**, $\Delta_r S^{\ominus}$, is the difference between the molar entropies of the pure, separated products and the pure, separated reactants, all substances being in their standard states.

## Checklist of equations

| Property | Equation | Comment | Equation number |
|---|---|---|---|
| Standard molar entropy from calorimetry | See eqn 3C.2 | Sum of contributions from $T = 0$ to temperature of interest | 3C.2 |
| Standard reaction entropy | $\Delta_r S^{\ominus} = \sum_{\text{Products}} \nu S_m^{\ominus} - \sum_{\text{Reactants}} \nu S_m^{\ominus}$ | $\nu$: (positive) stoichiometric coefficients; | 3C.3 |
| | $\Delta_r S^{\ominus} = \sum_J \nu_J S_m^{\ominus}(J)$ | $\nu_J$: (signed) stoichiometric numbers | |
| Temperature dependence of the standard reaction entropy | $\Delta_r S^{\ominus}(T_2) = \Delta_r S^{\ominus}(T_1) + \int_{T_1}^{T_2} (\Delta_r C_p^{\ominus}/T)\mathrm{d}T$ | | 3C.5a |
| | $\Delta_r S^{\ominus}(T_2) = \Delta_r S^{\ominus}(T_1) + \Delta_r C_p^{\ominus} \ln(T_2/T_1)$ | $\Delta_r C_p^{\ominus}$ independent of temperature | 3C.5c |